

Exercises: CRUD

You can check your solutions in the [Judge system](#).

1. Examine the Databases

Download and get familiar with the **SoftUni**, **Diablo** and **Geography** database schemas and tables. You will use them in the current and the following exercises to write queries.

Part I – Queries for SoftUni Database

2. Find All the Information About Departments

Create a SQL query that finds **all the available information about the Departments**.

Example

DepartmentID	Name	ManagerID
1	Engineering	12
2	Tool Design	4
3	Sales	273
...	...	...

3. Find all Department Names

Create a SQL query that finds **all Department names**.

Example

Name
Engineering
Tool Design
Sales
...

4. Find Salary of Each Employee

Create a SQL query that finds the **first name**, **last name** and **salary** for each employee.

Example

FirstName	LastName	Salary
Guy	Gilbert	12500.00
Kevin	Brown	13500.00
Roberto	Tamburello	43300.00
...	...	...

5. Find Full Name of Each Employee

Create a SQL query that finds the **first**, **middle** and **last name** for each employee.

Example

FirstName	MiddleName	LastName
Guy	R	Gilbert
Kevin	F	Brown
Roberto	NULL	Tamburello
...	...	...

6. Find Email Address of Each Employee

Create a SQL query that finds the **email address** of each employee by their **first and last name**. Consider that the email domain is **softuni.bg**. Emails should look like "**John.Doe@softuni.bg**". The **produced column** should be named "**Full Email Address**".

Example

Full Email Address
Guy.Gilbert@softuni.bg
Kevin.Brown@softuni.bg
Roberto.Tamburello@softuni.bg
...

7. Find All Different Employees' Salaries

Create a SQL query that finds **all different salaries of the employees**. Display the salaries only in one column, named "**Salary**".

Example

Salary
9000.00
9300.00
9500.00
...

8. Find All Information About Employees

Create a SQL query that finds **all information** about the employees whose **job title** is "**Sales Representative**".

Example

ID	First Name	Last Name	Middle Name	Job Title	Dept ID	Mngr ID	Hire Date	Salary	AddressID
275	Michael	Blythe	G	Sales Representative	3	268	...	23100.00	60
276	Linda	Mitchell	C	Sales Representative	3	268	...	23100.00	170

277	Jillian	Carson	NULL	Sales Representative	3	268	...	23100.00	61
...	...	...	...	...	...	...	...	...	...

9. Find Names of All Employees by Salary in Range

Create a SQL query to find the **first name**, **last name** and **job title** for all employees whose salary is in a **range between 20000 and 30000**.

Example

FirstName	LastName	JobTitle
Rob	Walters	Senior Tool Designer
Thierry	D'Hers	Tool Designer
JoLynn	Dobney	Production Supervisor
...	...	...

10. Find Names of All Employees

Create a SQL query that finds the **full name** of all employees whose **salary** is exactly **25000, 14000, 12500 or 23600**. The result should be displayed in a column, named "**Full Name**", which is a combination of the **first**, **middle** and **last** names, separated by a **single space**.

Example

Full Name
Guy R Gilbert
Thierry B D'Hers
JoLynn M Dobney

11. Find All Employees Without a Manager



Create a SQL query that finds the **first** and **last names** of those employees who **do not have a manager**.

Example

FirstName	LastName
Ken	Sanchez
Svetlin	Nakov
...	...

12. Find All Employees with a Salary More Than 50000



Create a SQL query that finds the **first name**, **last name** and **salary** for employees with a **salary** higher than **50000**. Order the result in decreasing order by salary.

Example

FirstName	LastName	Salary
Ken	Sanchez	125500.00
James	Hamilton	84100.00
...	...	...

13. Find 5 Best Paid Employees.

Create a SQL query that finds the **first** and **last** names of the **5 best-paid Employees**, ordered **descending** by their **salary**.

Example

FirstName	LastName
Ken	Sanchez
James	Hamilton
...	...

14. Find All Employees Except Marketing

Create a SQL query that finds the **first** and **last** names of all employees whose **department ID** is **not 4**.

Example

FirstName	LastName
Guy	Gilbert
Roberto	Tamburello
Rob	Walters

15. Sort Employees Table



Create a SQL query that sorts all the records in the **Employees** table by the following criteria:

- By **salary** in **decreasing** order
- Then by the **first name** **alphabetically**
- Then by the **last name** **descending**
- Then by **middle name** **alphabetically**

Example

ID	First Name	Last Name	Middle Name	Job Title	Dept ID	Mngr ID	Hire Date	Salary	AddressID
109	Ken	Sanchez	J	Chief Executive Officer	16	NULL	...	125500.00	177
148	James	Hamilton	R	Vice President of Production	7	109	...	84100.00	158
273	Brian	Welcker	S	Vice President of Sales	3	109	...	72100.00	134
...	...	...	...	...	...	...	...	...	...

16. Create View Employees with Salaries

Create a SQL query that creates a view "V_EmployeesSalaries" with **first name**, **last name** and **salary** for each employee.

Example

FirstName	LastName	Salary
Guy	Gilbert	12500.00
Kevin	Brown	13500.00
...	...	...

17. Create View Employees with Job Titles

Create a SQL query that creates a view "V_EmployeeNameJobTitle" with a **full employee name** and a **job title**. When the middle name is **NULL** replace it with an **empty string** ("").

Example

Full Name	Job Title
Guy R Gilbert	Production Technician
Kevin F Brown	Marketing Assistant
Roberto Tamburello	Engineering Manager
...	...

18. Distinct Job Titles

Create a SQL query that finds **all distinct job titles**.

Example

JobTitle
Accountant
Accounts Manager

Accounts Payable Specialist
...

19. Find First 10 Started Projects

Create a SQL query that finds the **first 10 projects** which were started, select **all the information** about them and **order** the result by **starting date**, then by **name**.

Example

ID	Name	Description	StartDate	EndDate
6	HL Road Frame	Research, design and development of HL Road ...	1998-05-02 00:00:00	2003-06-01 00:00:00
2	Cycling Cap	Research, design and development of C...	2001-06-01 00:00:00	2003-06-01 00:00:00
5	HL Mountain Frame	Research, design and development of HL M...	2001-06-01 00:00:00	2003-06-01 00:00:00
...	...	...	...	...

20. Last 7 Hired Employees

Create a SQL query that finds the **last 7 hired employees**, select their **first, last name** and **hire date**. **Order** the result by **hire date** descending.

Example

FirstName	LastName	HireDate
Rachel	Valdez	2005-07-01 00:00:00
Lynn	Tsoflias	2005-07-01 00:00:00
Syed	Abbas	2005-04-15 00:00:00
...	...	...

21. Increase Salaries

Create a SQL query that increases salaries by **12%** for all employees that work in one of the following departments – **Engineering, Tool Design, Marketing** or **Information Services**. As a result, select and display **only the "Salaries"** column from the **Employees** table. After this, you should restore the database to the original data.

Example

Salary
12500.00
15120.00
48496.00
33376.00
...

Part II – Queries for Geography Database

22. All Mountain Peaks

Display all the **mountain peaks** in **alphabetical order**.

Example

PeakName
Aconcagua
Banski Suhodol
Batashki Snezhnik
...

23. Biggest Countries by Population

Find the 30 biggest countries by population, located in **Europe**. Display the "**CountryName**" and "**Population**". **Order** the results **by population** (from biggest to smallest), **then by country alphabetically**.

Example

CountryName	Population
Russia	140702000
Germany	81802257
France	64768389
...	...

24. *Countries and Currency (Euro / Not Euro)



Find all the countries with information about their currency. Display the "**CountryName**", "**CountryCode**", and information about its "**Currency**": either "**Euro**" or "**Not Euro**". Sort the results by country name alphabetically.

Hint: Use **CASE ... WHEN**.

Example

CountryName	CountryCode	Currency
Afghanistan	AF	Not Euro
Åland	AX	Euro
Albania	AL	Not Euro
...	...	...

Part III – Queries for Diablo Database

25. All Diablo Characters



Display all **characters** in **alphabetical order**.

Example

Name
Amazon
Assassin
Barbarian
...